

Statistical & Quality

Body parts used

- **Pose confidence:** every body part's `_p` (pose probability) column; also frame-wise aggregates across **all** parts.
 - **Distribution tests:** whole-fish signals (**Bp_velocity_mean**, **Summed_movement**) built from **all body parts**.
 - **Fractal dimension:** a single anchor track (**bodypart16** position time series).
 - **Bout transition stats:** bout labels derived from whole-fish signals (**all body parts** indirectly).
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1) Pose confidence probabilities (data quality, per frame)

These summarize the DeepLabCut-style confidence (`{bp}_p`) across body parts, plus simple “low-confidence” flags.

- **Pose_p_mean**
Mean of `{bp}_p` across **all** body parts this frame.
Units: probability (0–1). Higher = better tracking this frame.
- **Pose_p_std**
Standard deviation of `{bp}_p` across parts (how uneven confidence is across the body).
Units: probability.
- **Pose_p_min, Pose_p_max**
Lowest / highest `{bp}_p` among body parts this frame.
Units: probability.
- **Low_conf_count_t01**
Number of body parts with `{bp}_p < 0.1` this frame.
Units: count (0 ... number of parts).
- **Low_conf_frac_t01**
Fraction of parts with `{bp}_p < 0.1` this frame.
Units: 0–1.

Rolling windows (trend over time):

- **Pose_p_mean_{W}, Pose_p_std_{W}, Low_conf_frac_t01_{W}**
Rolling mean/std or rolling mean low-confidence **fraction** over the last **W** frames.
Units: probability / probability / 0–1.

Body parts used: all parts' {bp}_p.

2) Distribution tests (normality, z-scores, surprise/entropy)

We test whether your main signals look Gaussian in a rolling window and provide standardized versions for outlier detection.

Signals tested per window **W**:

- **Bp_velocity_mean** ($\approx$ mm/s)
- **Summed_movement** (mm/frame)

Normality (Lilliefors / Kolmogorov–Smirnov for unknown mean/var):

- **Velocity_lilliefors_p_{W}, Movement_lilliefors_p_{W}**
P-values for “looks normal?” in the last **W** frames.
Units: probability (0–1); large p $\rightarrow$ **can't reject** normality.
- **Velocity_is_normal_{W}, Movement_is_normal_{W}**
1 if $p \geq 0.05$, else 0.
Units: binary.

Standardized (z-scored) series:

- **Bp_velocity_mean_z_{W}, Summed_movement_z_{W}**
Z-score within the last **W** frames (mean=0, std=1).
Use case: easy outlier thresholds like $|z| \geq 3$.

Distribution “spread” info:

- **Velocity_entropy_{W}, Movement_entropy_{W}**
Shannon entropy of the window's value histogram (broader = higher).
Units: bits.

Body parts used: all (indirectly, via the whole-fish signals above).

3) Fractal dimension (path complexity of a single anchor)

Box-counting-style fractal dimension of the 2D path traced by **bodypart16** over sliding windows.

Higher → more tortuous/space-filling; near 1 → nearly straight; approaching 2 → very space-filling.

- **Path_fractal_dimension_{W}**

Fractal (box-counting) dimension of the anchor's path over the last **W** frames.

Units: dimensionless (typically ~1.0–1.8 for animal tracks).

Body part used: **bodypart16** (tail anchor) positions (x, y).

4) Bout transition stats (how bouts start/stop; durations & intervals)

Built from the Active_swimming / Bout_ID labels (which come from whole-fish movement/velocity signals).

Per window **W**:

- **Inactive_to_active_prob_{W}**

Probability that a frame following **inactive** becomes **active** within the window.

Units: probability (0–1). “How likely am I to start a bout soon?”

- **Active_to_inactive_prob_{W}**

Probability that a frame following **active** becomes **inactive** within the window.

Units: probability (0–1). “How likely am I to end a bout soon?”

- **Bout_duration_mean_{W}, Bout_duration_median_{W}, Bout_duration_max_{W}**

Summary stats of active bout lengths seen inside the window.

Units: **seconds**.

- **Inter_bout_interval_mean_{W}**

Mean time between active bouts inside the window.

Units: **seconds**.

- **Bout_start_rate_{W}**

Starts per second (another view on frequency; complements Bout_frequency_{W} in Swimming Behavior).

Units: **Hz** (bouts/s).

Body parts used: **all** (indirectly, via Summed_movement / Bp_velocity_mean → bout labels).
